

**IN PROCESS QUALITY CONTROL AND FINISHED
PRODUCT QUALITY CONTROL TEST FOR
TABLETS AND CAPSULE**

SUBJECT: QUALITY CONTROL AND QUALITY ASSURANCE



SUBMITTED BY

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INTRODUCTION TO IPQC TESTING

Definition of IPQC

In-Process Quality Control (IPQC) refers to a branch of quality assurance that emphasizes continuous assessment and monitoring during the manufacturing process to ensure product quality. IPQC procedures are generally quick, simple, and rapid tests or inspections carried out during ongoing manufacturing operations.

It is a planned system to identify and monitor:

- Materials
- Equipment
- Manufacturing processes
- Operations

Finished Product Quality Control (FPQC) tests are performed on the finished product to confirm product quality before release to the market. IPQC consists of checks performed during manufacturing to ensure the product meets required quality standards before completion of the final product.

OBJECTIVES OF IPQC

The main objectives of IPQC are:

- To ensure uniformity and consistency during manufacturing
- To detect defects during the production stage

- To minimize product rejection and wastage
 - To maintain product quality and safety
 - To ensure compliance with pharmacopoeial standards
 - To reduce manufacturing errors and batch failures
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TABLETS

Introduction to Tablets

Tablets are solid unit dosage forms of medicaments prepared by compression or molding methods, with or without excipients, and intended for oral administration for local or systemic effects.

Tablets may vary in:

- Size
- Shape
- Weight

depending on the medicament and its mode of administration.

IN-PROCESS QUALITY CONTROL TESTS FOR TABLETS

The major IPQC tests performed for tablets include:

1. Hardness Test
2. Friability Test
3. Thickness Test

4. Weight Variation Test
 5. Content Uniformity Test
 6. Disintegration Test
 7. Dissolution Test
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1. HARDNESS TEST

Purpose

The hardness test is performed to measure the strength of a tablet and determine the force required to break it. It ensures that the tablet is strong enough to withstand handling, packaging, and transportation.

Procedure

1. Place one tablet between the hardness tester (Monsanto Hardness Tester).
2. Apply increasing force gradually.
3. Record the force at which the tablet breaks.
4. Repeat the procedure for multiple tablets.
5. Compare results with the specified product range.

Typical Range

- Normal tablets: **4–10 kp**
 - Chewable tablets: **>12 kp**
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2. THICKNESS TEST

Purpose

The thickness test ensures uniform tablet size and proper packaging. Variation in thickness should not exceed acceptable limits.

Procedure

1. Select 10 random tablets.
2. Measure thickness using a Vernier caliper or micrometer.
3. Record each value.
4. Calculate the average thickness.
5. Compare with standard specifications.

Acceptance Criteria

- Thickness variation should be within $\pm 5\%$ of the standard value.
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3. WEIGHT VARIATION TEST

Purpose

This test ensures that all tablets in a batch have uniform weight and contain the correct amount of drug.

Procedure

1. Select 20 tablets randomly.
2. Weigh each tablet individually using an analytical balance.
3. Calculate the average weight.
4. Compare each tablet weight with the average weight.

5. Determine compliance with pharmacopoeial limits.

Acceptance Limits

- $\pm 5\%$
- $\pm 7.5\%$
- $\pm 10\%$

depending on tablet weight.

4. DISINTEGRATION TEST

Purpose

The disintegration test determines the time required for tablets to break down into smaller particles under specified conditions.

Procedure

1. Prepare the USP disintegration apparatus.
2. Maintain medium temperature at $37 \pm 2^\circ\text{C}$.
3. Place 6 tablets in disintegration tubes.
4. Start the apparatus and timer.
5. Observe tablets until complete disintegration.
6. Record the disintegration time.

Acceptance Criteria

Tablet Type	Time Limit
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Uncoated Tablets	$\leq 15 \text{ min}$
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Tablet Type	Time Limit
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Coated Tablets	≤ 60 min
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Enteric-Coated Tablets	≤ 60 min
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Film-Coated Tablets	≤ 30 min
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Effervescent Tablets	≤ 5 min
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Soluble Tablets	≤ 3 min
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Dispersible Tablets	≤ 3 min
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5. DISSOLUTION TEST

Purpose

The dissolution test measures the rate and extent of drug release from the tablet dosage form.

Procedure

1. Select USP Apparatus I (Basket) or II (Paddle).
2. Fill dissolution vessel with appropriate medium.
3. Maintain temperature at $37 \pm 0.5^{\circ}\text{C}$.
4. Place tablets in vessels.
5. Start apparatus and timer.
6. Withdraw samples at specified intervals.
7. Replace withdrawn volume with fresh medium.

8. Analyze samples using:
 - HPLC
 - UV Spectrophotometry
9. Calculate percentage drug dissolved.
10. Compare with product specifications.

Common Dissolution Media

- Buffer pH 1.2
 - Buffer pH 4.5
 - Buffer pH 6.8
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CAPSULES

Introduction to Capsules

Capsules are solid unit dosage forms containing drug substances enclosed in hard or soft gelatin shells. Capsules may contain one or more active ingredients and are intended for oral administration.

Capsule shells are commonly filled with:

- Powders
 - Granules
 - Pellets
 - Beads
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IN-PROCESS QUALITY CONTROL TESTS FOR CAPSULES

Major IPQC tests for capsules include:

1. Weight Variation Test
 2. Content Uniformity Test
 3. Dissolution Test
 4. Disintegration Test
 5. Moisture Content Test
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1. WEIGHT VARIATION TEST FOR CAPSULES

Procedure

1. Select 10 gelatin capsules.
2. Weigh each capsule (Gross Weight).
3. Empty capsule contents completely.
4. Weigh the empty shell (Tare Weight).
5. Calculate Net Weight = Gross Weight – Tare Weight.
6. Calculate average net weight.
7. Compare individual net weights with the average.

Acceptance Criteria

If API ≥ 25 mg and constitutes $\geq 25\%$ of total weight:

- Not more than 2 capsules deviate by more than $\pm 10\%$
- None deviate by more than $\pm 20\%$

2. DISINTEGRATION TEST FOR CAPSULES

Procedure

1. Take 6 capsules.
2. Place capsules in the disintegration apparatus.
3. Use suitable medium:
 - Hard gelatin: purified water or 0.1 N HCl
 - Enteric-coated: 0.1 N HCl followed by phosphate buffer pH 6.8
4. Maintain temperature at **$37 \pm 2^{\circ}\text{C}$** .
5. Operate apparatus at **30 cycles/min**.
6. Observe complete disintegration.

Acceptance Criteria

- All 6 capsules should disintegrate within specified time.
- If failure occurs, test additional 12 capsules.
- At least 16 of 18 capsules must pass.

3. MOISTURE CONTENT TEST

Purpose

The moisture content test determines the amount of moisture present in capsule formulations to ensure stability and quality.

Procedure

1. Select representative capsule samples.
2. Weigh accurately (Initial Weight = W1).
3. Dry sample in oven at **105°C**.
4. Continue drying until constant weight.
5. Cool in desiccator.
6. Weigh dried sample (Final Weight = W2).
7. Calculate percentage loss on drying (LOD).

Formula

$$\text{LOD (\%)} = \frac{W1 - W2}{W1} \times 100$$

Acceptance Criteria

- Moisture content should comply with IP standards.

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